EDUCATION, WELL-BEING, AND EMPLOYMENT IN YOUNG ADULTHOOD

Predictive Evidence from the Panel Study of Income Dynamics

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Research Report

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Abstract

This study examines whether education and psychological well-being measured before 2023 improve prediction of employment in 2023 among young adults in the Panel Study of Income Dynamics Transition into Adulthood Supplement. The analytic sample contains 1,619 respondents with a valid employment outcome. I compare three logistic-regression specifications: a baseline model containing demographic, household-resource, and prior-employment measures; a second model adding education; and a third model adding PHQ-9 and GAD-7 symptom scores. On a stratified test sample, adding education raises ROC AUC from 0.717 to 0.741 and lowers the Brier score from 0.179 to 0.170. Adding well-being changes ROC AUC only from 0.741 to 0.742, although PR AUC rises from 0.863 to 0.869. Paired bootstrap intervals for the incremental changes all include zero, and the well-being model improves ROC AUC in only 14 of 30 repeated splits. A tuned XGBoost model does not outperform logistic regression. The results suggest that education and prior labor-market attachment are useful predictors of near-term employment, while PHQ-9 and GAD-7 scores provide little stable incremental predictive value once those factors are included. This finding concerns prediction, not the causal importance of mental health.

Keywords: young adulthood; employment; education; mental health; PSID; predictive modeling

1 Introduction

The transition from school to work is one of the central changes of young adulthood. Employment at this stage is connected to financial independence, skill accumulation, and later labor-market stability. Young adults enter this transition with different educational credentials, enrollment patterns, family resources, prior work histories, and psychological conditions. These

factors overlap: education can expand job opportunities, while anxiety and depressive symptoms may affect concentration, job search, attendance, and persistence.

This report asks a predictive question: how much do education and well-being measured before 2023 improve prediction of employment in 2023 beyond demographics, family economic resources, and prior employment? The distinction between prediction and causation matters. A variable can improve prediction without causing employment, and an important factor can add little predictive information when related variables are already included.

The analysis uses nested models so that the incremental contribution of education and well-being is visible. It evaluates discrimination, calibration, and class-specific performance rather than accuracy alone, and it checks stability with cross-validation, repeated splits, complete-case analysis, paired bootstrap resampling, and a comparison with XGBoost.

2 Literature and Analytical Framework

Human-capital theory provides a straightforward reason to expect education to predict employment. Schooling can build skills, signal readiness to employers, and expand access to occupations with formal credential requirements. At the same time, education is not randomly assigned. Family resources, prior achievement, health, and local opportunity affect both educational attainment and later employment, so the analysis does not interpret the education coefficients as causal returns to schooling.

Mental health may affect the school-to-work transition through several channels. Depressive and anxiety symptoms can reduce concentration, motivation, social interaction, and the consistency required for job search and work. Longitudinal evidence has linked persistent mental-health problems with later educational and employment disadvantage (Veldman et al., 2015). However,

symptom measures may overlap with information already contained in schooling and prior employment. The empirical question is therefore not whether mental health matters in general, but whether PHQ-9 and GAD-7 improve prediction after those observable histories are known. This framework leads to two expectations. First, education should improve out-of-sample prediction beyond the baseline because credentials and enrollment capture skills and current school-to-work position. Second, well-being may improve prediction, but its incremental contribution may be small if symptoms are correlated with education and prior labor-market attachment. The nested-model design tests these expectations without treating predictive improvement as evidence of causality.

3 Data and Study Design

The data come from the Panel Study of Income Dynamics (PSID) Transition into Adulthood Supplement. Predictors are primarily measured in 2021, with prior work and earnings measures referring to 2020, and the outcome is measured in 2023. This timing reduces direct overlap between predictors and outcome, but the design remains observational.

The linked sample contains 1,631 records. Twelve respondents have missing or ambiguous 2023 employment information, leaving 1,619 observations. Of these respondents, 1,163 (71.8%) are classified as working and 456 (28.2%) as not working. A stratified 80/20 split produces 1,295 training observations and 324 test observations with nearly identical employment rates.

Table 1 Analytic sample construction

Stage	N	Share retained
Linked 2021–2023 sample	1,631	100.0%
Valid 2023 employment outcome	1,619	99.3%
Training sample	1,295	80.0% of analytic sample
Test sample	324	20.0% of analytic sample
Complete cases on all predictors	1,398	85.7% of linked sample

4 Measures

4.1 Employment Outcome

The binary outcome equals one when a respondent reports working in 2023 and zero when a valid response indicates not working. Ambiguous or missing combinations are not forced into either category. This construction is appropriate for prediction but combines distinct labor-force states—such as unemployment, schooling, disability, and caregiving—within "not working."

4.2 Predictor Blocks

The baseline block includes sex, age, log family expenditure, 2021 employment status, weeks worked in 2020, log prior earnings, and an indicator for missing prior earnings. The education block contains highest completed education and enrollment status. The well-being block contains PHQ-9 depressive-symptom and GAD-7 anxiety-symptom scores.

PHQ-9 and GAD-7 items are recoded from values 1–4 to scores 0–3, and complete item responses are required for each total. Internal consistency is high (Cronbach's alpha 0.896 for PHQ-9 and 0.923 for GAD-7). Among valid responses, 34.4% have PHQ-9 scores of at least 10 and 29.0% have GAD-7 scores of at least 10. The scales are strongly correlated ($r = 0.743$), making their separate coefficients difficult to interpret.

5 Modeling Strategy

Each logistic model uses a pipeline fitted only on training data. Numeric variables receive median imputation and standardization. Categorical variables receive most-frequent imputation and one-hot encoding. Categories not observed during fitting are encoded as zeros at transformation. The three nested models are evaluated on the same held-out observations. Performance is measured with ROC AUC, precision-recall AUC (PR AUC), accuracy, balanced accuracy, log loss, and the Brier score. ROC AUC measures ranking; PR AUC is useful with an imbalanced outcome; balanced accuracy gives equal weight to both classes; and log loss and the Brier score evaluate probabilities. Lower log loss and Brier scores are better.

Additional checks include five-fold cross-validation, 30 repeated stratified splits, complete-case analysis, paired bootstrap intervals, calibration and subgroup checks, threshold analysis, and a tuned XGBoost classifier. These checks evaluate stability rather than search for the most favorable result.

The main comparison is incremental. Let M0 denote the baseline model, M1 the baseline plus education, and M2 the baseline plus education and well-being. The quantity of interest is the change in out-of-sample performance from M0 to M1 and from M1 to M2. I prefer the simplest model when additional complexity does not produce a stable improvement across metrics and resamples. This rule makes the final specification choice transparent rather than choosing a model only because it performs best on one split.

6 Results

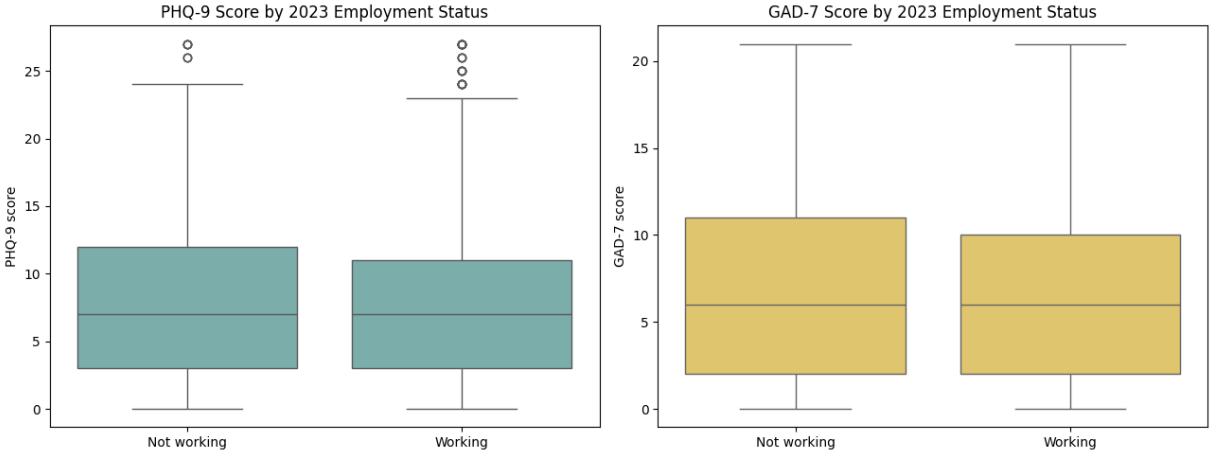
6.1 Descriptive Patterns

Table 2 Selected characteristics by 2023 employment status

Measure	Overall	Not working	Working
Female	54.7%	52.4%	55.5%
Age	21.39 (2.71)	20.76	21.64
Log family expenditure	10.71 (0.62)	10.62	10.74
Weeks worked in prior year	29.14 (22.44)	20.39	32.57
Log prior earnings	2.42 (3.59)	1.45	2.77
PHQ-9 score	7.92 (6.39)	7.96	7.90
GAD-7 score	6.80 (5.84)	7.06	6.70

Prior labor-market attachment differs more sharply by 2023 employment than average symptom scores do. Working respondents report about twelve more weeks of work in the prior year and higher prior earnings. Mean PHQ-9 scores are almost identical across outcome groups, and the GAD-7 difference is small.

Figure 1 PHQ-9 and GAD-7 scores by 2023 employment status



Source: Author's calculations using PSID Transition into Adulthood Supplement data.

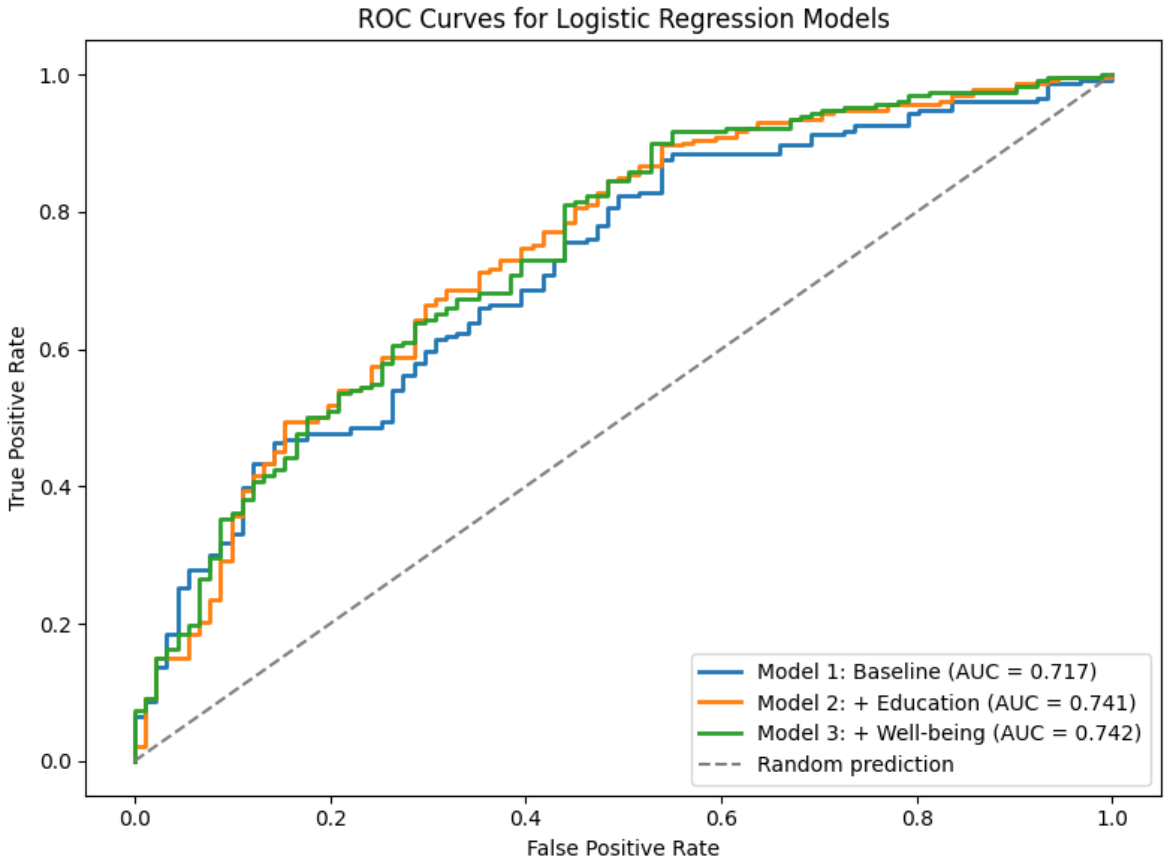
6.2 Nested Logistic Models

Table 3 Held-out test performance of nested logistic models

Model	ROC AUC	PR AUC	Accuracy	Balanced accuracy	Log loss	Brier
Baseline	0.717	0.862	0.731	0.572	0.534	0.179
+ Education	0.741	0.863	0.756	0.616	0.517	0.170
+ Well-being	0.742	0.869	0.765	0.629	0.514	0.169

Education produces the clearest improvement. Relative to baseline, it raises ROC AUC by 0.024, raises balanced accuracy by 0.044, and reduces the Brier score by 0.009. Adding PHQ-9 and GAD-7 then changes ROC AUC by only 0.001 and the Brier score by 0.001, while PR AUC improves by 0.006.

Figure 2 ROC curves for the nested logistic models



Source: Author's calculations using PSID Transition into Adulthood Supplement data.

At the 0.50 threshold, the full model correctly identifies 219 of 233 working respondents but only 29 of 91 not-working respondents. Recall is therefore 0.940 for working and 0.319 for not working. This asymmetry should not be hidden by overall accuracy of 0.765.

6.3 Incremental Value of Well-being

PHQ-9 alone, GAD-7 alone, and both scales together yield nearly identical test ROC AUC values (0.741, 0.742, and 0.742). In the full logistic model, standardized PHQ-9 and GAD-7

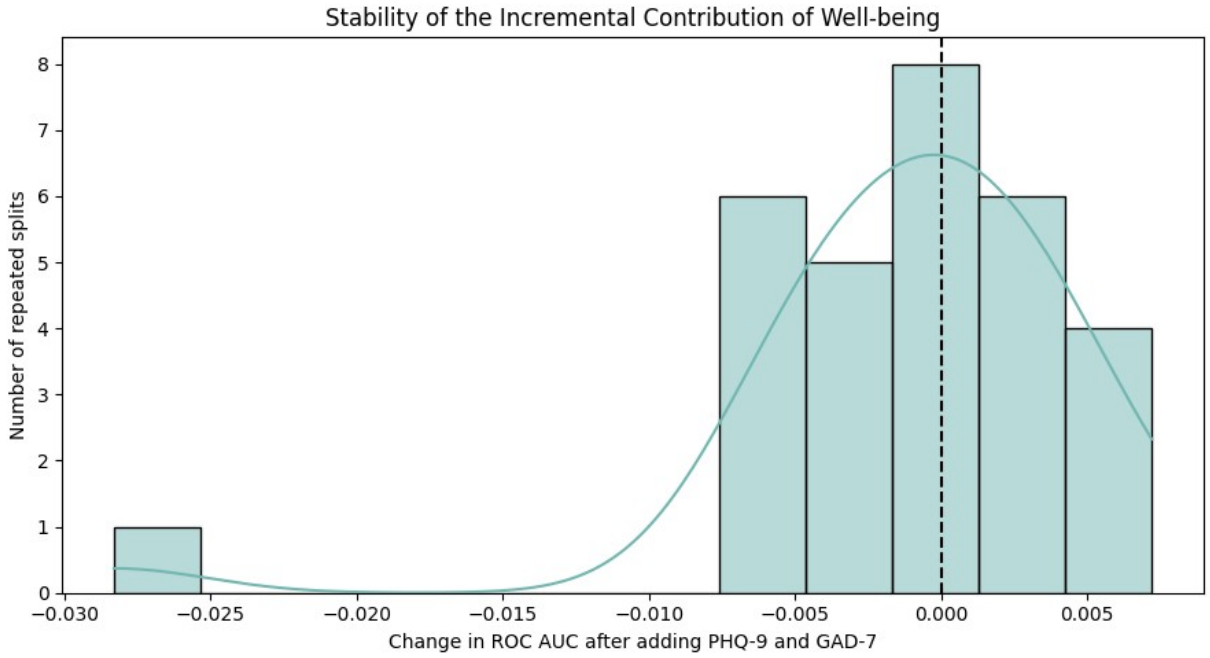
coefficients have opposite signs even though the scales are positively correlated. This is consistent with multicollinearity and does not justify separate causal interpretations.

Table 4 Paired bootstrap estimates for adding well-being

Metric	Education	Education + well-being	Mean change	95% interval
ROC AUC	0.7415	0.7419	+0.0005	−0.0117 to 0.0118
PR AUC	0.8628	0.8691	+0.0058	−0.0035 to 0.0170
Brier score	0.1699	0.1692	+0.0007 improvement	−0.0019 to 0.0032

Every bootstrap interval includes zero. Across 30 repeated splits, the full model has mean ROC AUC 0.728, compared with 0.729 for the education model. It improves ROC AUC in 14 of 30 splits and has a mean change of −0.001. The conclusion is not that mental health is unimportant; it is that these scales do not provide stable incremental prediction after education, resources, and prior employment are known.

Figure 3 Stability of the incremental ROC AUC contribution of well-being



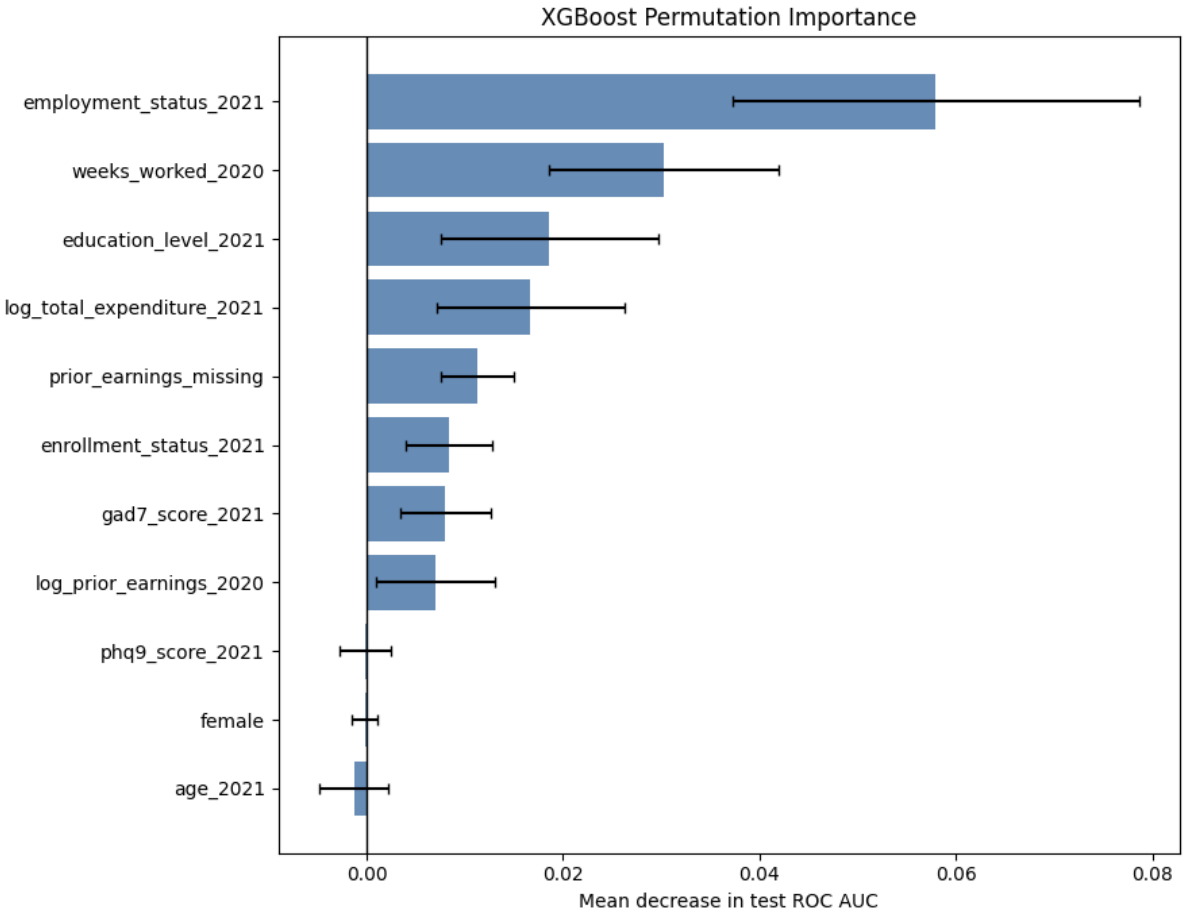
Source: Author's calculations using PSID Transition into Adulthood Supplement data.

6.4 XGBoost and Feature Importance

The tuned XGBoost model achieves test ROC AUC 0.735, PR AUC 0.869, balanced accuracy 0.565, and Brier score 0.177. Full logistic regression performs better on ROC AUC, balanced

accuracy, log loss, and Brier score while matching PR AUC. The more flexible algorithm therefore does not improve performance.

Figure 4 XGBoost permutation importance

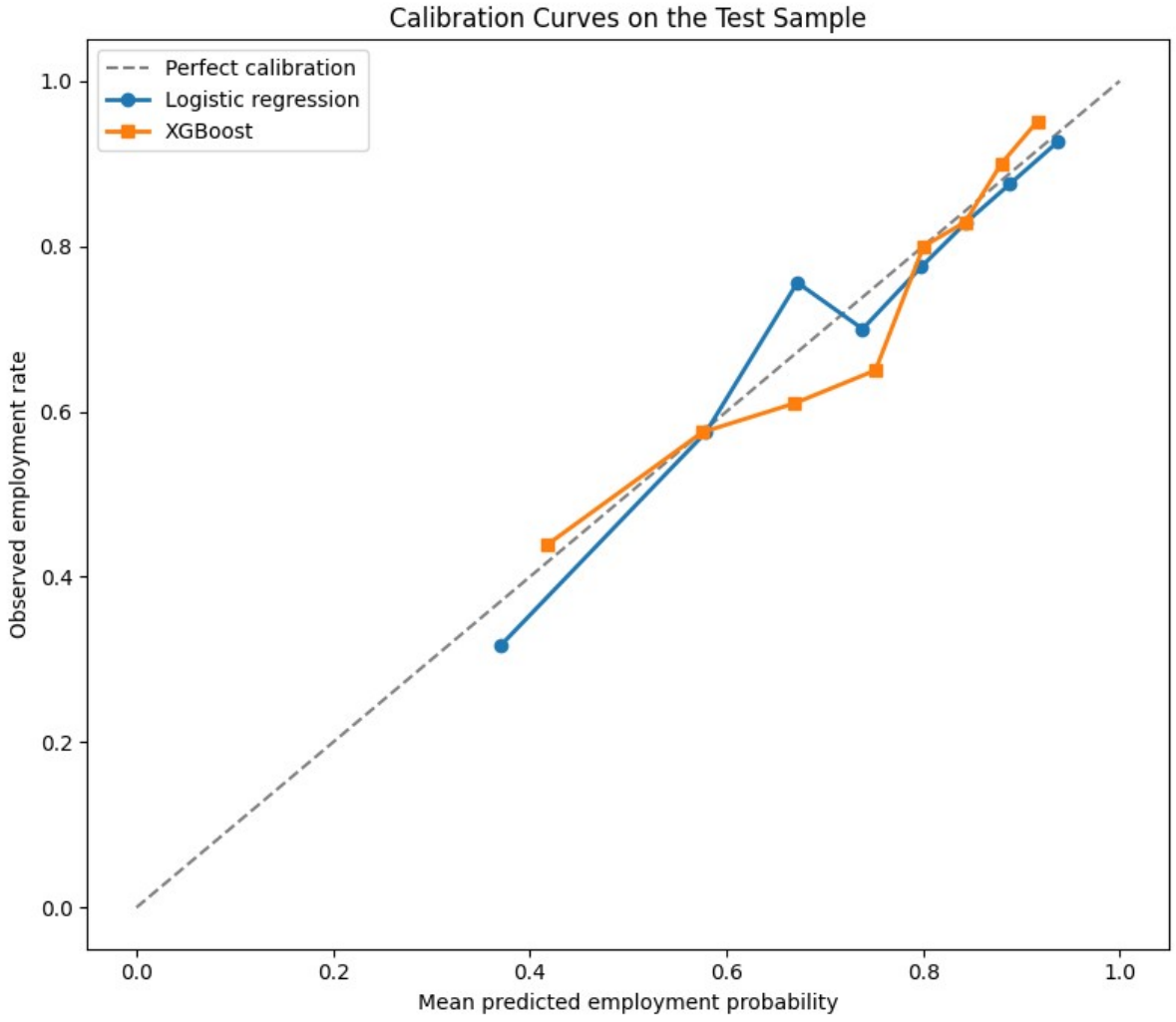


Source: Author's calculations using PSID Transition into Adulthood Supplement data.

Employment status in 2021 has the greatest permutation importance (mean ROC AUC decrease 0.0579), followed by weeks worked in 2020 (0.0303), education level (0.0186), and family expenditure (0.0167). GAD-7 has modest importance (0.0080), while PHQ-9 is effectively zero. These values measure model reliance, not causal effects.

6.5 Calibration and Thresholds

Figure 5 Calibration curves on the test sample



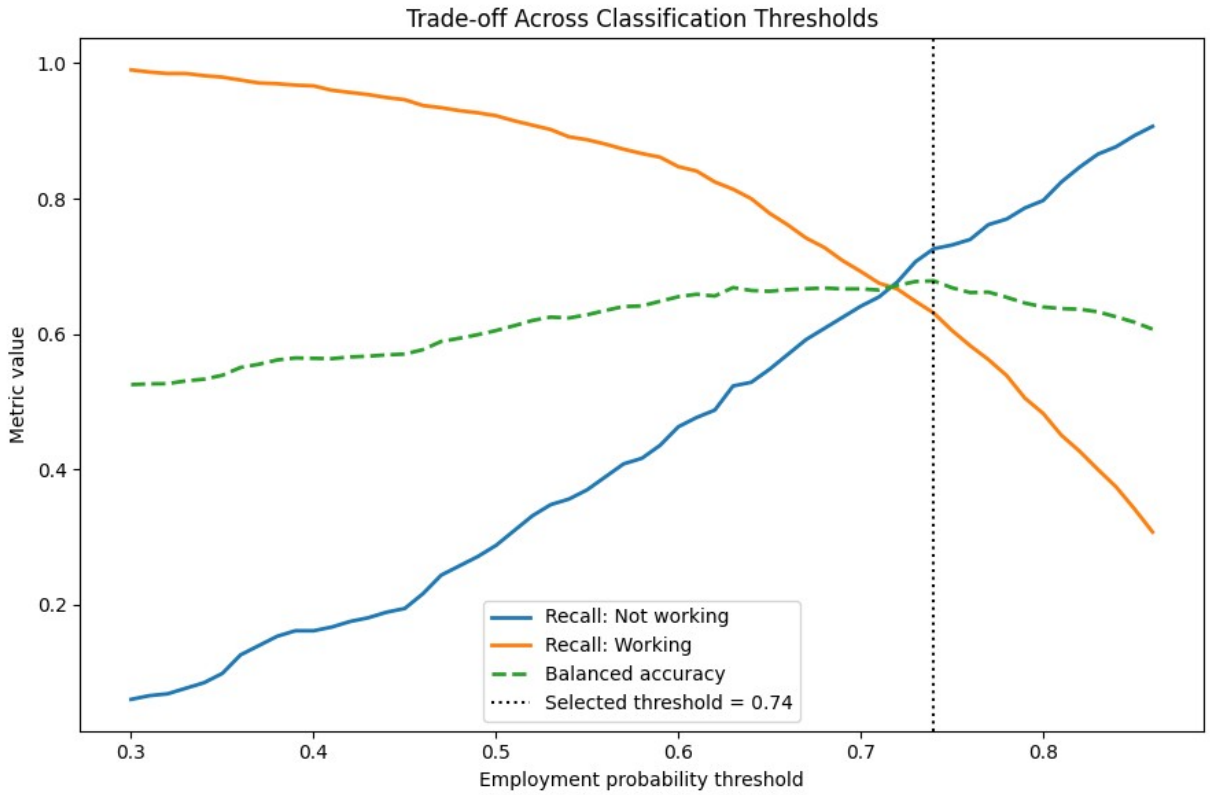
Source: Author's calculations using PSID Transition into Adulthood Supplement data.

The logistic model is reasonably calibrated across most probability groups and has a lower Brier score than XGBoost (0.169 versus 0.177). Some bins depart from the diagonal, so individual probabilities should still be interpreted cautiously.

Table 5 Consequences of two classification thresholds

Threshold	Accuracy	Balanced accuracy	Recall not working	Recall working	Precision not working
Default 0.50	0.765	0.629	0.319	0.940	0.674
Selected 0.74	0.660	0.670	0.692	0.648	0.434

Figure 6 Trade-off across employment-probability thresholds



Source: Author's calculations using PSID Transition into Adulthood Supplement data.

Raising the threshold to 0.74 nearly doubles recall for not working, but lowers recall for working and reduces total accuracy. The threshold is a policy choice, not a universally better setting. If identifying young adults at risk of not working is the priority, balanced accuracy may be more useful than total accuracy, but false alerts also matter.

6.6 Robustness and Subgroups

On the same 286 complete test cases, the imputation and complete-case models have nearly identical ROC AUC (0.750 and 0.751) and identical Brier scores (0.161). The imputation model has somewhat higher balanced accuracy (0.651 versus 0.632). This reduces concern that the conclusion is driven by missing-data treatment.

Subgroup estimates are descriptive because each test subgroup is small. ROC AUC is 0.771 for women and 0.708 for men; it is 0.750 for ages 17–21 and 0.731 for ages 22–26. These differences warrant monitoring but do not establish statistical significance or algorithmic bias.

7 Discussion and Policy Relevance

Recent labor-market attachment and education contain the most consistent predictive information. Prior employment status, weeks worked, and educational attainment summarize opportunities and constraints that often persist. Family expenditure adds information about economic context.

The well-being result is more nuanced. PHQ-9 and GAD-7 have high reliability and substantial symptom prevalence, yet add little after other variables are included. Symptoms may overlap with prior employment and education, and a single 2021 measurement cannot capture treatment, recovery, or shocks before 2023. Mental health may also matter more for hours, earnings, job quality, or retention than for a binary employment indicator.

For practical prediction, the education logistic model is a reasonable preferred specification because it is simpler and performs essentially as well as the full model across repeated splits. The full model remains useful for answering the incremental research question, but its small test-sample improvement is not a dependable gain.

From a public-management perspective, the threshold analysis is as important as the model ranking. A workforce program with limited staff cannot treat every predicted probability as an intervention. At the default threshold, the model misses many respondents who are not working. A higher threshold for classifying “working” identifies more of the not-working group, but it also generates more false alerts. An agency should therefore choose a threshold using program

capacity, the cost of outreach, and the cost of failing to reach someone who needs support—not accuracy alone.

The model should be used, if at all, as a screening aid rather than an eligibility rule. Prior employment and education are informative, but they may also reflect structural disadvantage. Automated exclusion based on these predictors could reproduce that disadvantage. A more defensible use would combine a calibrated risk score with human review, voluntary outreach, and routine monitoring of error rates by sex, age, race, and other legally and substantively relevant groups. The current subgroup analysis is too small to certify fairness.

The analysis also illustrates a business-analytics principle: additional data are valuable only when they improve a decision. PHQ-9 and GAD-7 may be important for service planning, but collecting sensitive health information solely to predict employment is difficult to justify when it adds almost no stable predictive value. The education model is therefore not only more parsimonious; it also reduces privacy and governance concerns.

8 Methodological Strengths and Limitations

The study has several strengths. Predictors precede the outcome, preprocessing is contained within the modeling pipeline, all nested specifications use the same test observations, and model quality is evaluated with ranking, probability, and class-sensitive metrics. Repeated splits, bootstrap intervals, complete-case comparison, calibration, and an alternative algorithm make it less likely that the conclusion depends on one favorable specification. Reporting a near-zero incremental result also avoids overselling.

This study does not identify causal effects. It does not apply PSID survey weights, so percentages and model performance describe the analytic sample rather than national prevalence.

The binary outcome combines different reasons for not working. Symptoms and employment are self-reported, and attrition may be selective. Some categorical levels are sparse. The test sample has only 324 respondents, making subgroup results uncertain. Finally, 2021–2023 spans an unusual post-pandemic period, and performance may not transport to other cohorts or labor markets.

9 Conclusion

Adding education to demographic, household-resource, and prior-employment predictors meaningfully improves prediction of 2023 employment. Adding PHQ-9 and GAD-7 produces only a very small and unstable incremental gain. Logistic regression performs at least as well as tuned XGBoost and provides better probability accuracy. The specific conclusion is that symptom scores, as measured here, do not materially improve near-term employment prediction once education and prior labor-market attachment are known. Future work should examine job quality and earnings, repeated symptom trajectories, survey-weighted estimates, and validation in later waves.

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Appendix Cross-validation and Reproducibility

Table A1 Five-fold cross-validation on the training sample

Model	Mean ROC AUC	SD	Mean PR AUC	Balanced accuracy	Log loss	Brier
Baseline	0.721	0.036	0.856	0.593	0.533	0.178
+ Education	0.734	0.031	0.864	0.604	0.523	0.174
+ Well-being	0.729	0.036	0.861	0.605	0.526	0.175

Continuous predictors were standardized within the pipeline; their odds ratios correspond to a one-standard-deviation increase. Categorical odds ratios are relative to an omitted reference level. The workflow does not estimate classical coefficient standard errors, so coefficients are treated as predictive associations rather than significance tests.